

The difference of two squares

The one product whose middle term cancels — and the fastest factorisation in algebra.

LESSON 100–101

2 × 40 MIN

MATEMATIKA 7, §45

S8 2.4

LESSON CLOCK

15'

25'

20'

25'

5'

The vanishing middle term	15 min
Factorising in one step	25 min
What does not work	20 min
Two uses	25 min
Homework	5 min

LEARNING OBJECTIVES

- Derive $(a - b)(a + b) = a^2 - b^2$ and explain why the middle term vanishes.
- Factorise a difference of two squares in one step.
- Recognise that a sum of two squares does not factorise this way.
- Use the formula for mental arithmetic and for cancelling fractions.

TEXTBOOK

National	pp. 290–296
Cambridge	Stage 8, pp. 25–30
Workbook	Exercise 2.4

01

Explanation

The vanishing middle term

$$(a - b)(a + b) = a^2 - b^2$$

STEP	WORKING
four products	$a^2 + ab - ba - b^2$
$+ab$	cancels $-ba$
what is left	$a^2 - b^2$

WHY THIS PAIR AND NO OTHER

The two middle products are equal and opposite only because one bracket has $+b$ and the other $-b$. Change either sign and the middle term returns.

Factorising in one step

Read the identity from right to left: any difference of two squares factorises immediately.

EXPRESSION	A	B	FACTORISED
$x^2 - 9$	x	3	$(x - 3)(x + 3)$
$4a^2 - 25$	$2a$	5	$(2a - 5)(2a + 5)$
$49 - y^2$	7	y	$(7 - y)(7 + y)$
$16x^2 - 81y^2$	$4x$	$9y$	$(4x - 9y)(4x + 9y)$

TAKE THE COMMON FACTOR OUT FIRST

$2x^2 - 18$ is not a difference of two squares as it stands. Take out the 2 first: $2(x^2 - 9) = 2(x - 3)(x + 3)$. Factorising is always finished, never stopped halfway.

What does not work

EXPRESSION	FACTORISES LIKE THIS?	WHY
$x^2 - 9$	yes	a difference of squares
$x^2 + 9$	no	a sum of squares
$x^2 - 5$	not over whole numbers	$\sqrt{5}$
$x^3 - 8$	not by this formula	cubes, not squares

$A^2 + B^2$ DOES NOT FACTORISE

Writing $x^2 + 9 = (x + 3)(x + 3)$ is wrong twice over: that product is $x^2 + 6x + 9$. A sum of two squares has no factorisation with real numbers at all.

Two uses

PRODUCT	AS A DIFFERENCE	ANSWER
$51 \cdot 49$	$50^2 - 1^2$	2499
$102 \cdot 98$	$100^2 - 2^2$	9996
$37 \cdot 43$	$40^2 - 3^2$	1591

And in a fraction the factorisation is what makes the cancelling possible:

$$\frac{x^2 - 9}{x + 3} = \frac{(x - 3)(x + 3)}{x + 3} = x - 3$$

THIS IS WHY FACTORISING COMES BEFORE FRACTIONS

Nothing can be cancelled from $x^2 - 9$ as it stands, because cancelling needs factors. Three lessons from now every algebraic fraction will begin with exactly this step.

02

Worked examples

EXAMPLE 1 Factorise $16x^2 - 81y^2$.

1

$\sqrt{16x^2} = 4x$

2

$\sqrt{81y^2} = 9y$

3

$= (4x - 9y)(4x + 9y)$

4

Check: the middle terms cancel ✓

ANSWER

$(4x - 9y)(4x + 9y)$

EXAMPLE 2 Factorise $2x^2 - 18$ completely.

① Common factor first: $2(x^2 - 9)$.

② $x^2 - 9 = (x - 3)(x + 3)$

③ $= 2(x - 3)(x + 3)$

ANSWER

$$2(x - 3)(x + 3)$$

EXAMPLE 3 Find $37 \cdot 43$ mentally.

① Both are 3 from 40.

② $37 \cdot 43 = (40 - 3)(40 + 3)$

③ $= 1600 - 9$

④ $= 1591$

ANSWER

$$1591$$

03 Interactive model

Put $51 \cdot 49$ and $50 \cdot 50$ side by side on the board and ask which is bigger; the difference of exactly 1 is the formula arriving before its name.

Factorise, then cancel

$$\frac{x^2 - 9}{x + 3}$$

① $x^2 - 9 = (x - 3)(x + 3)$

A
difference
of
two
squares.

② $\frac{(x - 3)(x + 3)}{x + 3}$

Now
there
is a
common
factor.

③ $x - 3$

Cancel
 $x +$
 $3 -$
and
note
 $x \neq$
 -3 .

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Difference of two squares	Kvadratlar ayirmasi	Разность квадратов
Sum of two squares	Kvadratlar yig'indisi	Сумма квадратов
Conjugate pair	Qo'shma ifodalar	Сопряжённые выражения
To cancel	Qisqartirish	Сокращать
Factorisation	Ko'paytuvchilarga ajratish	Разложение на множители
Square root	Kvadrat ildiz	Квадратный корень
One step	Bir qadamda	В один шаг
Identity	Ayniyat	Тождество

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$(a - b)(a + b)$ equals:

$x^2 - 9$ factorises to:

$x^2 + 9$ factorises to:

$4a^2 - 25$ factorises to:

$2x^2 - 18$ factorises completely to:

$51 \cdot 49$ equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x^2 - 9$

ANSWER $(x - 3)(x + 3)$

2. $x^2 - 16$

ANSWER $(x - 4)(x + 4)$

3. $a^2 - 1$

ANSWER $(a - 1)(a + 1)$

4. $49 - y^2$

ANSWER $(7 - y)(7 + y)$

5. $4a^2 - 25$

ANSWER $(2a - 5)(2a + 5)$

6. $(x - 2)(x + 2)$

ANSWER $x^2 - 4$

7. $51 \cdot 49$

ANSWER 2499

Medium · the standard question

7 PROBLEMS

1. $16x^2 - 81y^2$

ANSWER $(4x - 9y)(4x + 9y)$

2. $2x^2 - 18$

ANSWER $2(x - 3)(x + 3)$

3. $x^4 - 16$

ANSWER $(x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$

4. $102 \cdot 98$

ANSWER 9996

5. $37 \cdot 43$

ANSWER 1591

6.
$$\frac{x^2 - 9}{x + 3}$$

ANSWER $x - 3$

7. Does $x^2 + 25$ factorise?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. $9a^2 - 4b^2$

ANSWER $(3a - 2b)(3a + 2b)$

2. $(x + 1)^2 - 4$

ANSWER $(x - 1)(x + 3)$

3. $a^4 - b^4$

ANSWER $(a - b)(a + b)(a^2 + b^2)$

4.
$$\frac{4x^2 - 25}{2x + 5}$$

ANSWER $2x - 5$

5. $123^2 - 122^2$

ANSWER 245

6. Factorise $50^2 - 49^2$ and evaluate

ANSWER 99

7. $x^2 - \frac{1}{4}$

ANSWER $(x - \frac{1}{2})(x + \frac{1}{2})$

07 Homework**Homework — 5 tasks**

Always take out the common factor before looking for a difference of squares.

- Factorise $x^2 - 36$, $9a^2 - 16$ and $25 - 4y^2$.
- Factorise $3x^2 - 27$ completely.
- Find $61 \cdot 59$ and $104 \cdot 96$ by the formula.

4. Simplify $\frac{x^2 - 25}{x - 5}$.

5. Explain in one sentence why $x^2 + 16$ does not factorise.